

Zachary W. Walden

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Education

–Oklahoma Christian University

Bachelor of Science, Computer Engineering, GPA 4.0/4.0, Summa Cum Laude

Aug. 2018 - Apr. 2022

Honors & Awards

–Oklahoma Christian University

- Outstanding ECE Student (Sophomore, Junior & Senior years) *2020 - 2022*
- The President's Scholarship *2018 - 2022*
- The President's List *2018 - 2022*

Skills

Languages – C/C++ (Embedded(RTOS, PIC, AVR), Linux), Assembly (AVR), Verilog (RTL), CocoTb (Test benches), Bash, Python, MATLAB (Scripting)

Design – Kicad, Cadence Design Entry CIS, Digital (Microcontrollers, FPGA), Analog (Op-Amps, Transistors, Diodes)

Verification/Validation – Cadence PSpice, Oscilloscopes, Function Generators, Multimeters, Logic Analyzers, Network Analyzers

Prototyping – Breadboard, Iron & Hot Air, SMT (TSSOP, SOIC, & QFP packages), THT, Onshape CAD

Other – Docker, Git, General Linux, JSON, L^AT_EX, Agile

Experience

Home Caregiver –

May 2022 - Present

- Transported disabled individual to doctor's appointments, and for travel
- Ensured disabled individual was properly fed by procuring or cooking food
- Maintained disabled individual's home by doing recurring chores such as mowing the lawn and doing the dishes

–VisuALS Technology Solutions, LLC

2019 - Present

Open Source Developer –

May 2022 - Present

- Determined the cause of a display scaling issue in VisuALS' Android application, And determined what will be needed to solve the issue
- Researched running the application through WINE on Linux to provide a better development environment for open source developers
- Determined method to properly sign executable files so that windows automatically trusts the program upon installation

Machine Learning Developer –

Feb 2021 - Apr. 2022

- Improved training data creation rate by 900% by translating the program from Python to C++ and switching from png to bmp
- Used Nvidia Container Runtime in docker with Tensorflow GPU to accelerate model training by 100%
- Tweaked model architecture to allow compilation for Google's Edge TPU, accelerating inference performance
- Reduced eye detection error rate to nearly zero by writing a heuristics based filter

Java Programmer –

May 2019 - Aug. 2020

- Designed a Spring REST API to authenticate a software purchase using a tablet's IMEI number
- Connected the authentication API to a SQL database that stored the hash of the IMEI
- Designed a SQL database running on Google's Cloud
- Wrote dynamic Android U.I. elements
- Researched Machine Learning for text prediction (Tensorflow, Tensorflow Lite)

–Oklahoma Christian University

Jan. 2020 - Apr. 2022

Professor's Assistant - PCB Engineer –

Dec. 2020 - Jul. 2021

- Increased RGB laser system color capability by 20 times by designing a 3 channel modulating laser driver board
- Selected components to meet the frequency needs in excess of 12 MHz
- Designed, laid out and routed a 9V 3A switching buck converter
- Reverse engineered 7-color TTL driver board using a multimeter in continuity mode

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–Oklahoma Christian University

Jan. 2020 - Apr. 2022

Teaching Assistant: CENG-3203, CENG-3213, ENGR-1122 — *Jan. 2020 - Mar. 2020, Aug. 2021 - Apr. 2022*

- Helped students troubleshoot hardware and software issues with 8-bit AVR Microcontrollers, 74 series logic, HD44780 LCD's, Keypad encoders and parallel DACs in order to control a laser scanner system
- Instructed students on Assembly language concepts
- Guided students in constructing an autonomous robot driven by brushed DC motors controlled by a discrete MOSFET H Bridge
- Assisted students in debugging Analog & Digital circuitry, along with embedded C written for Microchip's 16-bit dsPIC line

Projects

–8-bit 5-stage Pipeline RISC CPU [View on GitHub](#)

Jan. 2022 - Apr. 2022

- Designed and implemented 32-bit fixed-length RISC instruction set comprising 42 instructions
- Architected and implemented, from scratch, an 8-bit bypassed CPU with a 5-stage integer pipeline in Verilog and ran it on a Xilinx Spartan 7 board
- Implemented and integrated a vga sync generator and a dumb framebuffer into the cpu using special instructions to write to the framebuffer
- Increased programming productivity 10 times by writing an assembler for my instruction set in Python
- Debugged modules using test benches written using the Cocotb framework for Python and viewing waveform dumps in GTKWave
- Used Xilinx's Integrated Logic Analyzer IP in conjunction with an internal reset trigger circuit to troubleshoot on chip behavior

–DC/DC Switching Boost Converter

Sep. 2020 - Nov. 2020

- Designed a discrete boost converter taking a 6V input and giving an adjustable 12V-24V output in steps of 0.1V using a PIC microcontroller for control
- Achieved an average of .54% error relative to the target output voltage
- Used PIC Datasheet and Family Reference Manual to write register level PWM and SPI drivers
- Wrote proportional-only control system with a bias adjustment to reach correct output voltage
- Designed and assembled a printed circuit board for the project
- Led a 3-person team, delegating responsibilities to members

–Network RGB LED Matrix [View on GitHub](#)

Apr. 2024 - Present

- Designed mounting system in Onshape for 12 3mm pitch 64x64 RGB LED matrices
- Built a server in C++ using [@hazeller/rpi-rgb-led-matrix](#) to take in frames over TCP sockets and display them on RGB LED matrices
- Modified [@hazeller/rpi-rgb-led-matrix](#) existing Nintendo DS emulator to send frame data over TCP sockets to a Network RGB LED Matrix server
- Used jsoncpp to read in a JSON configuration file
- Utilized pthreads and synchronization primitives to improve performance in both the server and client as well as safely shutting down separate threads
- Added LUT and color balance mechanisms to color and brightness match matrices from different manufacturers

–Stock Ticker [View on GitHub](#)

Dec. 2022 - Oct. 2023

- Used libcurl and jsoncpp to download and parse stock price data from ALPHAVANTAGE's stock API
- Wrote a graphics library to render characters and other primitives to a scene buffer
- Used [@hazeller/rpi-rgb-led-matrix](#) to render stock scenes to an RGB LED matrix
- Implemented a multithreaded architecture to ensure fetching stock data does not affect rendering

–Rubik's Cube Core Magnetization Mods

Aug. 2023 - Present

- Greatly increased puzzle stability by designing magnetic core to corner modifications for 6 different puzzles using Onshape, and manufactured them with a resin 3d printer

–RGB Laser Scanner Controller

Oct. 2020 - Apr. 2021

- Programmed a laser scanner control system in AVR assembly on an AtMega 32A
- Integrated a keypad and 2 line LCD into the system for entering characters and controlling display modes.
- The system can display up to 4 alphanumeric characters using the laser, and up to 16 using the LCD